

July 2025 CSE 220
Online on Signals and Their Properties
Subsections: A1, A2
Total Marks: 10
Total Time: 30 Minutes

January 7, 2026

In this assignment, you need to implement time sub-scaling of a sampled continuous-time signal in Python. You also need to plot the graphs of each of these signals.

Let the base signal be $x(t)$ (provided in the template file). You will compute and plot:

$$x(t), y(t) = x\left(\frac{t}{k}\right); \text{ Where } k \text{ is a positive integer}$$

Tasks

- Generate the time axis t and compute $x(t)$. (given)
- Implement a function `time_scale(...)` that produces $y(t) = x\left(\frac{t}{k}\right)$
- As the program is using a sampled continuous-time signal, time sub-scaling requires new samples between original values. For example $y(0) = x(0)$, $y(2) = x(1)$ but $y(1) = x(0.5)$ which is not available. So, you have to interpolate the value. You will implement a function `interpolate_signal(...)` to interpolate the missing sample values. Missing values will be calculated from the average of the nearest left and right values. $y(1) = 0.5 * (x(0) + x(1))$.
- Plot (with proper labels and legend) on the same figure: $x(t)$, $y(t)$. Ignore the values that go beyond the specified time range.

Marks Distribution

- Implementing interpolation: 4
- Implementing time sub-scaling: 4
- Plotting: 2